

RETAILSPHERE (PTY) LTD

Data Platform Programme

Non-Functional Requirements Document

Step 4 Deliverable — How the Platform Must Perform

Document ID	NFR-001
Version	1.0 Final
Date	15 May 2026
Author	Thozamile Mbalo – Data Solutions Architect
Reviewed by	Nomvula Mahlangu – Chief Data Officer
Total Requirements	32 requirements across 8 categories
Status	Draft – Pending CDO Sign-off

1. Purpose

This document defines the non-functional requirements for RetailSphere Data Platform. It answers one question – how must the platform perform? Where FRD-001 defined what the platform must do, this document defines the standards the platform must meet in terms of performance, scalability, availability, security, data quality, and usability.

Requirements in this document are derived from the stakeholder interviews documented in BPA-001, functional requirements in FRD-001, and applicable South African legislation including POPIA and the Companies Act 71 of 2008.

2. Scope

This document covers non-functional requirements only – how the platform must perform, scale, stay available, remain secure, comply with legislation, retain data, maintain quality, and serve business users effectively. Functional requirements – what the platform must do are documented separately in FRD-001.

Code	Category	Requirements
NFR-PER	Performance	NFR-PER-001 to NFR-PER-007
NFR-SCA	Scalability	NFR-SCA-001 to NFR-SCA-004
NFR-AVL	Availability	NFR-AVL-001 to NFR-AVL-003
NFR-SEC	Security	NFR-SEC-001 to NFR-SEC-004
NFR-LEG	Legal/Compliance	NFR-LEG-001 to NFR-LEG-003
NFR-RET	Data Retention	NFR-RET-001 to NFR-RET-003
NFR-DQA	Data Quality	NFR-DQA-001 to NFR-DQA-005
NFR-USB	Usability	NFR-USB-001 to NFR-USB-003

3. Requirement Format and Conventions

Each non-functional requirement is documented using standard five-field format. Requirements are grouped into eight categories. Where information cannot yet be confirmed pending stakeholder or legal input, requirements are marked as TBC with clear action and owner recorded.

Field	Description
Requirement ID	Unique identifier in format NFR-[CAT]-[NUMBER]
Category	Which of the eight requirement categories
Requirement	The standard the platform must meet
Rationale	Why this standard exists and which stakeholder or regulation drives it
Measure	How is compliance is tested – pass or fail condition
Priority	Critical / High / Medium / Low

4. Performance

Performance requirements define how fast the platform must respond and how quickly processes must complete. Seven requirements are defined covering dashboard load time, pipeline completion, query response, alert latency, opt-out propagation, concurrent users, and daily data volume.

NFR-PER-001 — Dashboard Load Time

Requirement ID	NFR-PER-001
Category	Performance
Requirement	The platform must maintain dashboard load time within 5 seconds
Rationale	Slow dashboard load times delays business decisions, specifically procurement savings and supplier delivery performance.
Measure	The dashboard loads within 5 seconds under normal load conditions. Tested with 20 concurrent users. Measured from click to full render.
Priority	Critical

NFR-PER-002 — Pipeline Completion Time

Requirement ID	NFR-PER-002
Category	Performance
Requirement	The overnight pipeline must complete all layers – Bronze, Silver, and Gold by 5 am to ensure we still have 2 hours for troubleshooting and fixing issues.
Rationale	Pipeline not completed by 5 am will cause delays for Johan if the issue is not resolved within 2 hours, which is why we created this 2-hour buffer to allow time to intervene before Johan notices anything.
Measure	Pipeline completed by 5 am and any retries completed within 2 hours.
Priority	Critical

NFR-PER-003 — Query Response Time

Requirement ID	NFR-PER-003
Category	Performance
Requirement	Query response time within 3 seconds for single metric queries.
Rationale	Delayed response time leads to delayed decisions across all domains and that affects the whole business.
Measure	Single metric queries return results within 3 seconds under normal load. Tested with 30 concurrent users.
Priority	Critical

NFR-PER-004 — Backdated Posting Alert Latency

Requirement ID	NFR-PER-004
Category	Performance
Requirement	Backdated posting after nightly extraction must be detected and an alert sent within one hour.
Rationale	When the backdated posting alert is not sent within one hour, Johan has nothing to approve and we risk having incorrect finance numbers, which could cost Johan his reputation and career.
Measure	Detected backdated posting – a notification alert sent within one hour.
Priority	Critical

NFR-PER-005 — Opt-out Propagation

Requirement ID	NFR-PER-005
Category	Performance
Requirement	Opt-out from Salesforce CRM and propagation update all three marketing systems within 4 hours.
Rationale	If this is missed there could be legal and financial implications to the business that could lead to loss of revenue and reputation.
Measure	When a customer opts-out all 3 marketing systems are updated and excluded from all campaigns within 4 hours.

Priority	Critical
----------	----------

NFR-PER-006 — Concurrent Users

Requirement ID	NFR-PER-006
Category	Performance
Requirement	TBC – Information required.
Rationale	We need the concurrent user count beforehand as if we guess we might under provision – when the system hits peak it will crash and cost business money. Over provisioned we might be over spending on infrastructure we barely utilise.
Measure	TBC
Priority	High
Action	Consult domain owners to confirm.
Owner	T. Mbalo

NFR-PER-007 — Daily Data Volume

Requirement ID	NFR-PER-007
Category	Performance
Requirement	Estimated ~300MB data per day for the Bronze Layer and ~50MB for the Gold layer.
Rationale	This could impact the chosen infrastructure and budget.
Measure	Integration process from the source is producing ~300MB per day into the Bronze Layer. After cleaning and transformation into the Gold layer we see ~50MB per day.
Priority	Medium

5. Scalability

Scalability requirements define how the platform must handle growth over time without requiring redesign or causing performance degradation. Four requirements are defined covering store growth, member volume growth, new source system onboarding, and data volume growth over time.

NFR-SCA-001 — Store Growth

Requirement ID	NFR-SCA-001
Category	Scalability
Requirement	The platform must support the growth from 87 stores up to 200 stores without redesigning or performance degradation.
Rationale	The platform would experience performance degradation and congestion if we do not design the Bronze layer to handle store growth – this could bring down the platform and affect all domains.
Measure	The data platform can still work as expected when we add an extra 113 stores with similar data requirements without delays.
Priority	High

NFR-SCA-002 — Member Volume Growth

Requirement ID	NFR-SCA-002
Category	Scalability
Requirement	The platform should support the growth from 2.1 million loyal members to 5 million loyal members, as this gives us the buffer to avoid redesign later.

Rationale	The Contact History fact table will be most affected as it will be a large table. The Gold layer should support this growth otherwise all reports dependent on this table will experience latency.
Measure	The Bronze layer and the Gold layer Contact History fact table are able to handle growth to 5 million loyal members without performance degradation.
Priority	High

NFR-SCA-003 — New Source System Onboarding

Requirement ID	NFR-SCA-003
Category	Scalability
Requirement	Onboard a new source system within 4 weeks to allow for unforeseen delays, otherwise achievable in 2 weeks, without disrupting existing pipelines.
Rationale	Careless onboarding of a new source system could disrupt existing pipelines that feed current reports across all six domains.
Measure	Added a new source system and loaded ~300MB of data into the Bronze layer, completed all transformation through Silver layer, loaded into Gold layer, tested metrics and signed off.
Priority	Medium

NFR-SCA-004 — Data Volume Growth Over Time

Requirement ID	NFR-SCA-004
Category	Scalability
Requirement	The platform must be able to process ~600MB per day without infrastructure redesign.
Rationale	Failing to process 600MB data per day could delay the platform and lead to delayed decisions across the business.
Measure	Processed 600MB of data and the platform handles it without disrupting the current pipeline.
Priority	High

6. Availability

Availability requirements define how reliable and recoverable the platform must be. Three requirements are defined covering platform uptime, recovery time objective, and recovery point objective.

NFR-AVL-001 — Platform Uptime

Requirement ID	NFR-AVL-001
Category	Availability
Requirement	The platform must maintain 99.9% uptime – equivalent to a maximum of 8.76 hours downtime per year and approximately 45 minutes per month.
Rationale	Failing to keep uptime at 99.9% will lead to lost revenue as no functions could operate while the system is down and that will be across the business.
Measure	Testing recovery processes and planned maintenance confirms we meet the 8.76 hours downtime per year and ~45 minutes per month standard.
Priority	Critical

NFR-AVL-002 — Recovery Time Objective

Requirement ID	NFR-AVL-002
Category	Availability

Requirement	The platform must be up and running within 2 hours after a failure.
Rationale	This would lead to delayed reports causing delayed decisions across the business, especially impacting Johan's 7am requirement.
Measure	Testing confirms the platform is up and running within 2 hours after a failure.
Priority	Critical

NFR-AVL-003 — Recovery Point Objective

Requirement ID	NFR-AVL-003
Category	Availability
Requirement	The platform can only afford to lose 30 minutes of data, requiring backups every 15 minutes to cover near real-time recovery.
Rationale	Failed backups within 30 minutes could lead to loss of revenue, legal actions, and business reputation damage.
Measure	Tested full backups and 15-minute log backups confirming the database can be recovered to the point-in-time of failure and all backups are valid.
Priority	Critical

7. Security

Security requirements define how the platform protects data and controls access. Four requirements are defined covering role-based access control, data encryption, audit logging, and authentication.

NFR-SEC-001 — Role-based Access Control

Requirement ID	NFR-SEC-001
Category	Security
Requirement	The platform must make use of role-based access control only. No unauthorised principals – users or applications allowed.
Rationale	Unauthorised principals on the platform could lead to data breaches, legal consequences, and loss of business reputation and revenue.
Measure	Tested each domain user confirming they can only see their own domain data using user-defined database roles and security groups.
Priority	Critical

NFR-SEC-002 — Data Encryption

Requirement ID	NFR-SEC-002
Category	Security
Requirement	The platform must encrypt data both at rest and in transit.
Rationale	Failure to encrypt data at rest and in transit would lead to loss of revenue and legal consequences as we must adhere to POPIA obligations and, for global expansion, GDPR.
Measure	Created a test account – no personal data is visible as it is masked using data masking. All pipeline transformations and loads use encryption to protect data in transit.
Priority	Critical

NFR-SEC-003 — Audit Logging

Requirement ID	NFR-SEC-003
Category	Security
Requirement	The platform must audit all data access, user login attempts successful and failed, when a user was granted permissions, and approved by whom.

Rationale	Failure to audit data access means changes happen silently with no one accountable, leading to data loss, data breaches, and loss of revenue.
Measure	Tested login attempts – can see login time, IP Address, and hostname. Login failures captured with number of attempts and start and end times. All permission requests logged via ticketing system and approved by domain owners or managers before actioned by DBAs.
Priority	Critical

NFR-SEC-004 — Authentication

Requirement ID	NFR-SEC-004
Category	Security
Requirement	The platform must employ both multi-factor authentication and Single Sign-On as both are industry standards.
Rationale	Without proper authentication we could compromise company systems leading to data breaches, loss of revenue, and reputational damage.
Measure	Tested Single Sign-On – logged in from laptop and accessed reports without signing in again. Tested multi-factor authentication – approval sent to phone confirmed.
Priority	Critical

8. Legal and Compliance

Legal and compliance requirements define how the platform meets South African legislative obligations. Three requirements are defined covering POPIA compliance, King IV and Companies Act compliance, and data subject rights.

NFR-LEG-001 — POPIA Compliance

Requirement ID	NFR-LEG-001
Category	Legal and Compliance
Requirement	The platform must meet POPIA obligations - opt-out propagation within 4 hours, right to erasure, and data subject access requests.
Rationale	Failure to comply would lead to legal consequences, loss of trust, revenue, and reputation - something RetailSphere cannot afford.
Measure	Tested opt-out with a test account — Salesforce CRM, Marketing Cloud, Meta, and GA4 all updated within 4 hours. Tested loyalty member data deletion with confirmation notification sent. Data subject access request fulfilled via encrypted CSV or PDF using member ID as password.
Priority	Critical

NFR-LEG-002 — King IV and Companies Act Compliance

Requirement ID	NFR-LEG-002
Category	Legal and Compliance
Requirement	The platform must comply with King IV and the Companies Act by adhering to data encryption, access control, audit trails, data retention, and incident response requirements.
Rationale	Failure to comply could lead to data breaches, loss of business, deregistration from the JSE, legal consequences, and loss of 2.1 million loyal members through lack of trust.
Measure	Tested login failures and successes. Audit trail of tickets logged and approved by line managers for data access. Database changes tracked through source control and CI/CD platform (TBC — platform choice pending). Data encrypted at rest using TDE and in transit using Always Encrypted. PII masked using column data masking.
Priority	Critical

NFR-LEG-003 — Data Subject Rights

Requirement ID	NFR-LEG-003
Category	Legal and Compliance
Requirement	The platform must be able to update a loyalty member's personal details when requested without losing history. Members can request correction or deletion of inaccurate or outdated personal information.
Rationale	Failure to comply would lead to legal consequences as data subjects under POPIA have the right to correct or delete data and to be informed of such changes.
Measure	Tested loyalty member detail updates using SCD Type 2 — both old and new versions intact and no financial reports missing any data.
Priority	Critical

9. Data Retention

Data retention requirements define how long different categories of data must be kept. Three requirements are defined covering financial data retention, personal data retention, and operational data retention. Two requirements are currently TBC pending legal and business sign-off.

NFR-RET-001 — Financial Data Retention

Requirement ID	NFR-RET-001
Category	Data Retention
Requirement	The platform must retain financial data for a minimum of 7 years.
Rationale	Failure to retain financial data for 7 years will lead to legal consequences including fines and is required by the Companies Act.
Measure	Tested archiving processes with dummy data backdated 8 years. Alert set up to flag data at 6.5 years old — giving 6 months to retest the archive process before the 7-year deletion window.
Priority	Critical

NFR-RET-002 — Personal Data Retention

Requirement ID	NFR-RET-002
Category	Data Retention
Requirement	TBC — Requires legal sign-off
Rationale	POPIA states personal data should only be kept as long as necessary. The business must define how long loyalty member personal data is retained. Requires sign-off from Priya Naidoo and Rethabile Sithole.
Measure	TBC
Priority	High

NFR-RET-003 — Operational Data Retention

Requirement ID	NFR-RET-003
Category	Data Retention
Requirement	TBC — Requires business sign-off
Rationale	Not yet confirmed how far back transaction history should go. Rethabile Sithole must advise and sign off as this impacts CLV calculation and segment history.
Measure	TBC
Priority	Medium

10. Data Quality Standards

Data quality requirements define the standards the platform must enforce across all five dimensions of data quality. Five requirements are defined covering completeness, accuracy, consistency, timeliness, and uniqueness.

NFR-DQA-001 — Completeness Standard

Requirement ID	NFR-DQA-001
Category	Data Quality Standards
Requirement	The platform must not accept any incomplete records, especially critical columns specified by each domain. The platform must fail the process and send an alert notification to be investigated by the engineer, and resume data flow from the point of failure once corrected.
Rationale	Missing records could lead to incomplete information for business decisions, forfeiting the purpose of the platform.
Measure	Tested NOT NULL constraints on loyalty member ID number, GRN delivery date, and invoice supplier code — error thrown when data is missing in critical columns.
Priority	High

NFR-DQA-002 — Accuracy Standard

Requirement ID	NFR-DQA-002
Category	Data Quality Standards
Requirement	The platform must validate data against source systems and the Bronze layer to confirm they are equal. Any variance pauses the pipeline and sends a detailed alert. Data flow resumes from point of failure — not a full reload.
Rationale	Inaccurate data leads to wrong decisions, someone losing their job, and loss of revenue.
Measure	Tested validation steps between source systems and Bronze layer — matching records pass, non-matching records trigger detailed alert. Engineer investigates and resumes from point of failure.
Priority	High

NFR-DQA-003 — Consistency Standard

Requirement ID	NFR-DQA-003
Category	Data Quality Standards
Requirement	The platform must enforce data consistency through data synchronisation and validation rules between source systems and Bronze layer, ensuring updated data is reflected across all systems.
Rationale	Failure to enforce consistency leads to domain misalignment and legal consequences, especially the POPIA obligation to unsubscribe members from all campaigns within 4 hours.
Measure	Tested opt-out propagation — after unsubscribing from Salesforce CRM an alert confirms Marketing Cloud, Meta, GA4, and the platform are all updated with event details, member identity, and reason.
Priority	Critical

NFR-DQA-004 — Timeliness Standard

Requirement ID	NFR-DQA-004
Category	Data Quality Standards
Requirement	The platform must enforce data timeliness by understanding each domain's report delivery expectations. A 40% buffer alert fires at 60% of the delivery window — if a

	pipeline is expected in 6 hours, an alert fires at 3.6 hours to flag a potential delay before the deadline is breached.
Rationale	Unexpected delays impact decisions at specified thresholds, leading to legal consequences and loss of revenue.
Measure	TBC — testing criteria provided by stakeholders after delivery thresholds are confirmed.
Priority	High

NFR-DQA-005 — Uniqueness Standard

Requirement ID	NFR-DQA-005
Category	Data Quality Standards
Requirement	The platform must enforce data uniqueness by ensuring no unintended duplicate records exist to maintain data integrity and accuracy.
Rationale	Duplicate records lead to misreporting where the same record is counted multiple times, skewing results and causing wrong decisions and loss of revenue.
Measure	Tested unique ID generation rules during loading from Bronze to Silver — Bronze data kept as-is for root cause investigation. Surrogate key introduced in Customer dimension generated in Silver layer confirming one person across three systems.
Priority	High

11. Usability

Usability requirements define how the platform must serve business users who are not technically trained. Three requirements are defined covering self-service reporting, dashboard design standards, and alert readability.

NFR-USB-001 — Self-service Reporting

Requirement ID	NFR-USB-001
Category	Usability
Requirement	The platform must enable self-service reporting. Business users are not required to write code or contact the data team for existing reports. All calculations done in Gold layer. Full column name descriptions — no abbreviations in business-facing reports. New metric requests go through change management.
Rationale	Failure to implement self-service reporting introduces confusion and wasted time as business users would need to learn SQL, leading to delayed decisions and impacting business outcomes.
Measure	TBC — Sipho and Rethabile to test that they can see their metrics and reports without confusion or coding.
Priority	High

NFR-USB-002 — Dashboard Design Standards

Requirement ID	NFR-USB-002
Category	Usability
Requirement	All reports and dashboards must be uniform using colours (Navy, Black, White, Light Blue, Gold), font (Calibri), and layout (most important metrics at top) to ensure every dashboard looks and feels consistent.
Rationale	Inconsistent dashboards make it difficult for business users to navigate — feeling like they are learning something new every time, leading to frustrated users who avoid the platform.

Measure	Verify all reports and dashboards follow design standards before going live and being handed to business users.
Priority	High

NFR-USB-003 — Alert Readability

Requirement ID	NFR-USB-003
Category	Usability
Requirement	All alerts must be detailed and readable for business users. Must contain a clear heading, descriptive reason, priority level (Critical, High, Medium, Low, Info), time occurred, and action steps required.
Rationale	Alerts without sufficient detail will be ignored or missed — causing silent failures leading to loss of revenue, legal consequences, and loss of loyal members.
Measure	Each alert notification verified — heading is clear, reason is descriptive, time is visible, action steps are clear.
Priority	Critical

12. Next Steps

- CDO review and sign-off on NFR-001 v1.0.
- Resolve TBC requirements — NFR-PER-006 concurrent users, NFR-RET-002 personal data retention, NFR-RET-003 operational data retention, and NFR-DQA-004 timeliness thresholds — before Step 6 sign-off.
- NFR-RET-002 legal review initiated with Priya Naidoo and Rethabile Sithole.
- NFR-RET-003 business sign-off obtained from Rethabile Sithole.
- CI/CD platform decision made and NFR-LEG-002 measure updated accordingly.
- Proceed to Step 5 — Source System Investigation.
- Upload NFR-001 to GitHub: docs/04-non-functional-requirements/ NFR-001_v1.0.docx

13. Document Change Log

Version	Date	Author	Description
1.0	15 May 2026	T. Mbalo	Initial release. 32 requirements across 8 categories. All requirements derived from BPA-001 stakeholder interviews and applicable South African legislation.

— End of Document —